

PGEE 2019

April 2019

1 Paper Format

- General Aptitude: 45 questions (2 marks each)
- Computer Science: 60 questions 30 of 1 mark each. Another 30 of 2 marks each
- **NO Multiple answer questions, NO negative marking**

2 Aptitude

A poll was conducted, whose results were as below:

- 12 people said that the paper was easy and they were able to solve most questions.
- 81 people said that the paper was medium and with some efforts, they were able to solve most questions.
- 42 people said that the paper was difficult and they had to skip some questions due to it.

Overall, the paper was medium to difficult, compared to CS, which was much easier.

3 CS

Nothing was asked from TOC and Compilers. Other than, prepare for the exam as if you would prepare for GATE, and not by the syllabus given in the website.

1. Worst case time complexity for insertion in binary tree. A: $O(n)$
2. No. of networks in class C addressing A: 2^{21}
3. Data structure used in iterative DFS: A: Stack
4. Algorithm used to find MST. A: Kruskal

5. Number of child processes using `fork()`, when run n times: A: 2^{n-1}
6. Number of edges in spanning tree: $n - 1$
7. Which representation finds if there is edge between two vertices in $O(1)$ time? Ans: Adjacency matrix
8.


```

      int cnt = 0;
      recursive_count(int n){
        if (n==0){
          return n;
          cnt++;
          recursive_count(n/10);
        }
      }
      int main(){
        recursive_count(123456790);
        printf("%d", cnt);
        return 0;
      }
      
```
9. Trace of matrix where $A[i][j] = i+j$ A: 6
10. The time complexity to delete an arbitrary node from binary heap. A: $O(n)$
11. <https://www.geeksforgeeks.org/gate-gate-cs-2000-question-21/>
12. One question from microprocessor
13. <https://gateoverflow.in/2202/gate2010-23>
14. In a token ring network the transmission speed is 10^7 bps and the propagation speed is 200 metres/micro second. The 1-bit delay in this network is equivalent to:
15. Static variables are stored in: Heap, Stack, Stack and Heap, None
16. <https://www.geeksforgeeks.org/gate-gate-cs-2012-question-8/>
17. Pipelining increases the: Latency, Throughput and two other options A: Improves throughput
18. $(A * B')(A + B) = ?$. A' B' are complements of A and B. A: 0
19. For i goes from 1 to 100

```

if (i % 3 == 0)
printf("Great)
if(i%5 == 0)
printf("india")
printf("\n-----");

```

Count number of times GreatIndia is printed.

A: 6 times

20. Cycle frequency is 2GHZ. What is clock time?

A: 0.5 ns

21. <https://www.geeksforgeeks.org/gate-gate-cs-2010-question-24/>

22. $X(X + YZ) = ?$ A: X

23. One question was there where they initialized the variables to 0 and were calculating product ans to that was 0. A: 0

24. Max height of binary tree with 14 nodes? A: 14 (counting height as starting from one)

25. XOR gate can be formed using which gates ?

26. Binary representation of -77 in 2's compliment

27. Decimal representation of 250.2 which is in base 8

28. Which DS can be used to access initially added elements in $O(1)$ time: stack, queues, linked list or heap ? A: Queue

29. High level to assembly language converter?

30. Finding strongly connected components using DFS. A: Refer to Dasgupta, Vazirani et al.

31. <https://www.geeksforgeeks.org/gate-gate-cs-2006-question-20/>

32. <https://www.geeksforgeeks.org/gate-gate-cs-2007-question-65/>

33. <https://gateoverflow.in/490/gate2008-67>

34. Three-four cache questions, which were basic.

35. One question on C++ classes.

36. One C++ program whose gist was, it was rotating the array clockwise in every iteration and we had to print the output after three iterations.

37. `for(int i = 0; i < n-1; i++){`
 `for(j = i+1; j < n; j++){`
 `if(A[i] + A[j] == x) cout << Whatever`

Basically, doing a linear scan of the array and finding pairs who summed up to a particular number x.

38. Two questions from subnetting.
39. <https://gateoverflow.in/43470/gate2009-58>
40. Number of cells in an eight variable K-map A: 256
41. One question on CRC polynomial